

Assignment 1

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June 20, 2026

Problem 1

First let's define the two random variables, and their respective probability distribution. For the random variable U , the number of tries before the first success can be expressed from a geometric distribution which takes two parameters: k the number of trials, and p the probability of success. Thus,

$$U \sim \text{Geometric}(k, p) \quad (1)$$

For the random variable V , the number of trials needed to get 2 heads can be expressed as a negative binomial distribution which takes three parameters: r the number of success, x the number of failure, and p the probability of success per trial. Thus

$$V \sim \text{NBinom}(r, x, p) \quad (2)$$

From the definition of the problem, we know that $S = \{\text{head}, \text{tail}\}$, and

$$P(\omega) = \begin{cases} \frac{1}{2} & \text{if } \omega = \text{head} \\ \frac{1}{2} & \text{if } \omega = \text{tail} \end{cases} \quad (3)$$

where p is the probability of obtaining head.

```
# Set the seed
set.seed(48109)
# Number of simulations
sims <- 1000
# Number of trials
trials <- 100
# Probability of success
p <- 0.5
# Define empty vector for U
u <- c()
```

```

# Define empty vector for V
v <- c()

# Initiate the loop
for (i in 1:sims) {
  heads <- 0
  trials <- 0
  first_head <- NA

  while (heads < 2) {
    trials <- trials + 1
    flip <- rbinom(1, 1, p)
    if (flip == 1) {
      heads <- heads + 1
      if (heads == 1) first_head <- trials
    }
  }
  u[i] <- first_head
  v[i] <- trials
}

# Plotting the output
plot(
  jitter(u) ~ jitter(v),
  xlab = "Number of trials to get 2 heads",
  ylab = "Number of trials before 1st head"
)
abline(lm(u ~ v), col = "red")

# Testing the correlation
cor(u, v, method = "spearman")

```

```
[1] 0.7178861
```

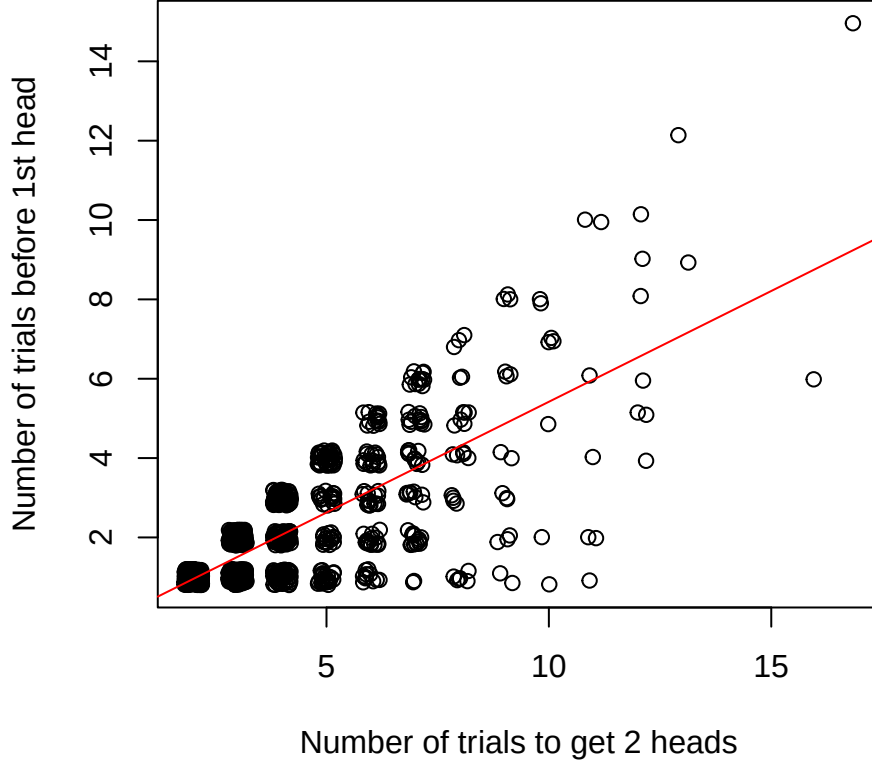


Figure 1: Joint distribution of the random variables U and V

From Figure 1, we can see that the two random variables are positively correlated. The Spearman correlation also confirm this intuition as the correlation is 0.71.

Problem 2

Part A

In this problem, I have a $1/3$ probability of choosing the door that contains the car. Once I pick the door, Monty Hall will truthfully tell me which door conceals the car with a probability of $1/2$, and will lie to me with a probability of $1/2$.

Now, in this problem, we want to know what's the probability that the car is behind the door I choose conditional on Monty telling me it is. Let $A \equiv$ Car is behind the door I choose and $B \equiv$ Monty tells me the car is behind the door I choose, we can write the quantity of interest using Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (4)$$

Where by the law of total probability, we can rewrite $P(B)$ as

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

where $A_i \in \{A, A^c\}$. Now, it gets a little bit more complicated. I don't know whether or not Monty tells the truth or if he's lying. But, what I do know is that if he's lying or not is independent of my selection as it is the result of flipping a fair coin that is made regardless of the door I picked. When Monty tells me that the car is behind the door I choose, he could be either lying or telling the truth, with each a probability of $1/2$. As I had an initial probability of $1/3$ of picking the correct door, the joint probability of picking the correct door and Monty telling me truthfully that the car is behind the door I selected is equal to the compound probability of each, which is $1/6$. This is justified as the door I picked is independent on whether or not Monty has to tell me truth. The joint probability of two independent events is equal to the product of each separate probability.

Now, the case that Monty is lying when he tells me the car is behind the door I selected. If he's lying, it implies that I picked a door with a goat behind. I had an initial probability of $2/3$ of picking a goat, and Monty is lying with probability $1/2$. However, when Monty is lying, he could have either picked my door or the other one containing the goat to lie about. This implies that there was a $1/2$ change he would have picked my door when he's lying. We can then compute the probability that he's lying when he tells me the car is behind my door: $1/3 \times 1/2 \times 1/2 = 1/6$.

Now, we can compute the posterior with the above information:

$$P(A|B) = \frac{\frac{1}{6}}{\frac{1}{3}} = \frac{1}{2} \quad (5)$$

We can also make a simulation

```
set.seed(48109)

sims <- 1000

car_behind_door <- rbinom(sims, 1, prob = 1 / 3)
lying <- rbinom(sims, 1, prob = 0.5)
monty_picks_door <- rbinom(sims, 1, prob = 0.5)

B <- (car_behind_door == 1 & lying == 0) |
     (car_behind_door == 0 & lying == 1 & monty_picks_door == 1)

mean(car_behind_door[B])

[1] 0.5339806
```

Part B

Let's define the following $A \equiv$ Car is behind door M and $B \equiv$ Monty says car is behind door M , where door M is not the door I initially picked. To switch door, the probability of the car being behind door M given that Monty says so should be greater than the probability of me picking the correct door when Monty says that the car is behind door M

$$P(A|B) > P(\text{My door}|B) \quad (6)$$

Let's start by finding $P(A|B)$. We can compute this by using Bayes rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (7)$$

The numerator can be understood as the probability of Monty telling the truth when the car is behind door M , which is $1/6$. About the denominator, this quantity can be expressed using the law of total probability

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) \quad (8)$$

where A^c is that the car is not behind door M . In this case, there are two possibilities as there are 2 doors that contain goats. Thus we have $2 \cdot 1/12$ for these cases. Hence,

$$P(B) = \frac{1}{6} + \frac{1}{12} + \frac{1}{12} = \frac{1}{3} \quad (9)$$

Thus, we have

$$P(A|B) = \frac{1}{6} \cdot \frac{3}{1} = \frac{1}{2} \quad (10)$$

In the case of $P(\text{My door}|B)$, we would have the following

$$P(\text{My door}|B) = \frac{P(B|\text{My Door})P(\text{My door})}{P(B)} \quad (11)$$

Where the numerator is the joint probability of the car being behind my door and Monty saying it's behind door M , which is $1/12$. Thus

$$P(\text{My door}|B) = \frac{1}{12} \cdot \frac{3}{1} = \frac{1}{4} \quad (12)$$

From the above two results, we see that we are better off to change to door M when Monty says the car is behind door M .

```

set.seed(48109)
sims <- 1000

# 1 = my door, 2 = door M, 3 = third door
car_location <- sample(1:3, sims, replace = TRUE)

lying <- rbinom(sims, 1, prob = 0.5)

monty_says <- numeric(sims)

for (i in 1:sims) {
  if (lying[i] == 0) {
    monty_says[i] <- car_location[i]
  } else {
    goat_doors <- setdiff(1:3, car_location[i])
    monty_says[i] <- sample(goat_doors, 1)
  }
}

mean(car_location[monty_says == 1] == 1)

[1] 0.5015198

mean(car_location[monty_says == 2] == 2)

[1] 0.5259939

```